

Algebra 1
Unit 1
Lesson 3 Practice

Name **Answer Key** _____
Date _____
Period _____

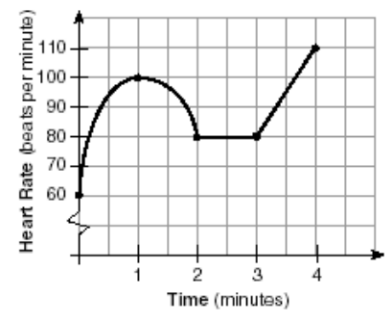
The function $G(m) = 49.5m + 89.75$ represents the cost of joining the gym in addition to the one time membership fee. The cost, G , is measured in dollars for m months.

- Find the value of $G(6)$.
 $49.5(6) + 89.75$
 386.75
- What does $G(6)$ represent in the context of the problem?
6 months of gym membership costs \$386.75
- Is the value of $G(9.75)$ reasonable to interpret in the context of the problem?
No, customers pay for full months of membership, not partial.

A function $H(m)$ is modeled in the graph below. It shows the heart rate of a swimmer during a 4 minute race where m is the number of minutes and H is the heart rate of the swimmer.

- Find the approximate value of the graph for $H(1.5)$.

$$H(1.5) \approx 96$$



- What does $H(1.5)$ mean in the context of the problem?
After 1.5 minutes, the heart rate is 96 BPM.
- After how many seconds is the swimmers heart rate 80 beats per minute?
 ≈ 10 sec
- Why does it make sense that $H(2) > H(0)$ in this context?
After 2 minutes of swimming, the heart rate is higher than before they started swimming.

The table below represents the temperature of a cup of coffee as it begins to cool. Temperature readings are taken every 5 minutes and measured in degrees Fahrenheit ($^{\circ}\text{F}$).

Minutes, m	0	5	10	15
Temperature, $T(m)$	197	186	170	164

8. Explain the meaning of $T(0) = 197$.
The initial temperature of the coffee is 197° .
9. Determine the value of $T(10)$.
 170°
10. Interpret $T(10)$ for the given context.
After 10 minutes of cooling, the temperature of the coffee is 170° .